

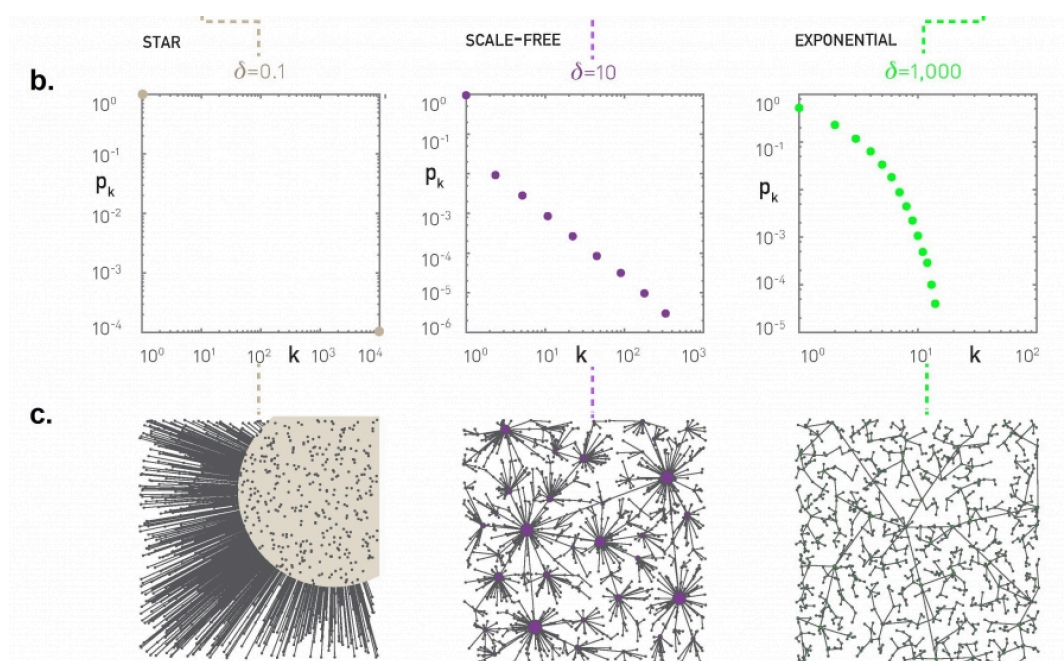
L12: Optimization-based Network Formation Model (cont')

The previous model (and some variants of it) has been studied extensively. The mathematical analysis of the model is quite lengthy, however, at least for our purposes, and so let us focus on numerical results and some basic insight.

First, what happens when δ is so low that the distance term does not matter? The maximum possible distance between two nodes in a unit square is $\sqrt{2}$. If $\delta < 1/\sqrt{2}$, we have that $\delta d_{i,j} < 1$. So, for any new node, it is cheaper to connect directly to the center (which has $h_j = 0$) rather than any node with $h_j \geq 1$. In other words, when $\delta < 0.707$, the resulting network has a hub-and-spoke topology because every node connects directly to the center node (see leftmost plots in the visualization – $N=10,000$ nodes).

On other hand, what happens if δ is so large that the distance term dominates over the path length h_j ? When we have N random points on a unit square, the average distance between a point and its nearest neighbor scales with $1/\sqrt{N}$ (do you see why?). The path length to the central node, on the other hand, scales slowly with N (logarithmically with N – recall that the network is a tree). So if $\delta \gg \sqrt{N}$, we have that $\delta d_{i,j}$ dominates over the h_j term. In that case, a new node- i will choose to connect to the nearest node- j . The resulting network in this case has an exponential degree distribution, as it is highly unlikely that a node will be the nearest neighbor to many other nodes.

When δ is between these two extremes ($1/\sqrt{2}$ and $\sqrt{N}$), the model produces topologies with a strong presence of hubs. The hubs connect to many other nearby nodes (i.e., the cost of those connections is quite low) and additionally the few hubs connect either directly to the center node or indirectly, through another hub node, to the center (i.e., the hubs have low h_j value). This hierarchical structure allows almost all nodes in the network to have a very low cost value C_j . The resulting degree distribution can be a power-law – even though the exact shape of the distribution depends on the value of N and δ .



We have seen several models so far that are very different with each other -- but they can all create power-law degree distributions. *What does this mean?*

It is possible that very different mechanisms (e.g., preferential attachment, link-copy, optimization of cost versus path-length, etc) can all create the same network statistics -- in terms of the degree distribution or even other properties (clustering, diameter, etc). *So we should be careful to not jump into conclusions about the mechanism that has generated a given network, just because that network exhibits a certain statistical property.*

Food For Thought

If you are mathematically inclined, you can study the following paper that presents a rigorous analysis of a variant of the previous model:

"Heuristically Optimized Trade-Offs: A New Paradigm for Power Laws in the Internet" 

(https://link.springer.com/chapter/10.1007/3-540-45465-9_11), by Alex Fabrikant, Elias Koutsoupias and Christos H. Papadimitriou.